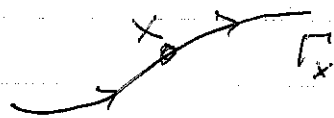


α - and ω -limit sets for dynamical systems

$\uparrow$ $\uparrow$

$t \rightarrow -\infty$ $t \rightarrow +\infty$



asymptotic behavior
limit sets associated
with trajectories as
 $t \rightarrow \pm\infty$

If $y \in \omega(x)$, then $\exists t_k \rightarrow \infty$ s.t.
 $\lim_{k \rightarrow \infty} \varphi_{t_k}(x) = y$

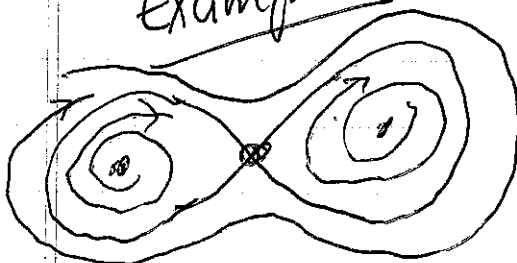
$t_1 < t_2 < t_3 < \dots$
 $\dots t_{-2} < t_{-1} < 0 < t_1 < \dots$

If $y \in \alpha(x)$, then $\lim_{k \rightarrow \infty} \varphi_{t_k}(x) = y$

Note: if ~~then~~ $z \in \Gamma_x$, then $\omega(z) = \omega(x)$
so we can write $\omega(\Gamma_x)$, and $\alpha(\Gamma_x)$
 $= \omega(x)$ $= \alpha(x)$

If x^* is an equilibrium then $\omega(x^*) = \alpha(x^*) = x^*$

Example:



ω -limit sets:

3 fixed pt.

each ~~interaction~~ homoclinic loop

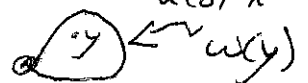
figure 8

$z \rightarrow \infty \leftarrow \omega(z)$



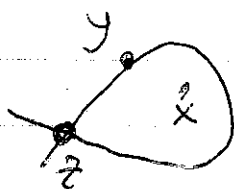
$\omega(y) = x^*$

$\alpha(y) = x^*$



Properties of ω -limit sets (Proved in text)

1. Existence: ω -limit set of a bounded orbit is non-empty
2. Closure: ω -limit set is closed
3. Invariance: If $x \in \omega(x_0)$, then $\phi_t(x) \in \omega(x_0) \quad \forall t > 0$
4. Connectedness: ω -limit set of bounded orbit is connected.
5. Transitivity: If $z \in \omega(y)$, $y \in \omega(x)$, then $z \in \omega(x)$



Poincaré-Bendixson Thm (Proved in text): For planar vector fields, $\dot{x} = V(x)$, $x \in \mathbb{R}^2$, that generate $\phi_t(x)$. If equilibria are isolated and $\phi_t(x)$ is bounded for $t > 0$, then either

1. $\omega(x)$ is an equilibrium
2. $\omega(x)$ is a periodic orbit
- or 3. for each $u \in \omega(x)$, $\alpha(u)$ & $\omega(u)$ are equilibria

Lecture 8 p.3

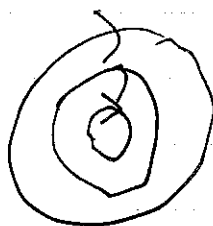
(1)

(2)

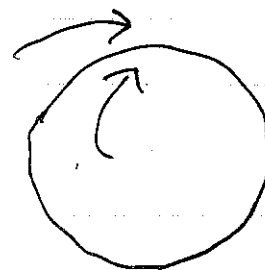
(3)
homoclinic orbit

heteroclinic orbit

Limit cycles: a periodic orbit γ that is the ω - or α -limit set of a pt. $x \notin \gamma$



vs.



not a limit cycle

limit cycle is an isolated periodic orbit

Stability of limit cycles - 2 approaches to be compared. ~~But today I present just one~~

$\dot{x} = V(x)$, $x \in \mathbb{R}^n$, V is C^k , ($k \geq 1$)
limit cycle $\gamma(t) = \gamma(t+T)$, $T > 0$ is (minimal) period.

Approach 1: linear stability analysis

let $x(t) = \gamma(t) + y(t)$ $\dot{\gamma} = V(\gamma(t))$

$$\dot{x} = \dot{\gamma} + \dot{y} = \underline{V(\gamma)} + \dot{y}$$

$$= V(\gamma + y) = \underline{V(\gamma)} + D_{\gamma} V(\gamma(t)) y + \text{...}$$

linear stability problem:

$$(*) \quad \dot{y} = \underbrace{D_{\gamma} V(\gamma(t))}_{= A(t) = A(t+T)} y \quad \left. \vphantom{\dot{y}} \right\} \text{Floquet Theory setting}$$

can case

Fundamental matrix soln ^{of (*)} $\Phi(t)$:

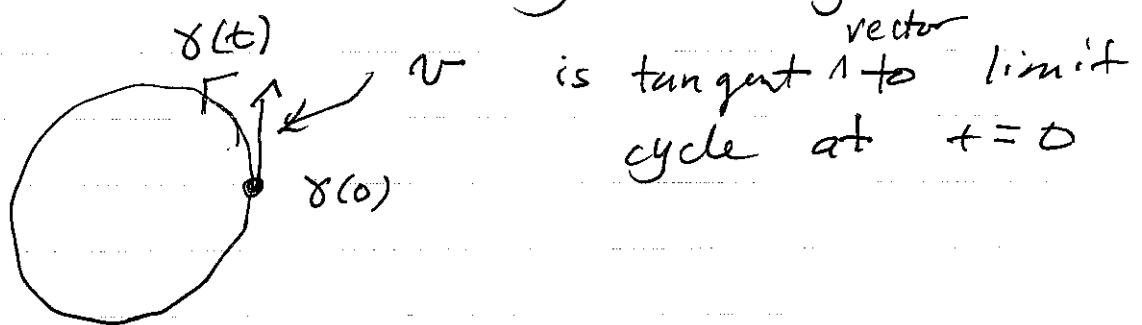
$$\dot{\Phi} = A(t) \Phi, \quad \Phi(0) = \text{Identity matrix}$$

if $y(0) = y_0$, then $y(t) = \Phi(t) y_0$, $y(T) = M y_0$
 $M = \Phi(T)$
 = monodromy matrix

$$y_0 \xrightarrow{\Phi_T} M y_0 \xrightarrow{\Phi_T} M^2 y_0 \xrightarrow{\Phi_T} \dots M^k y_0$$

Before we discussed that if all eigenvalues μ of M satisfy $|\mu| < 1$ then linear asymptotic stability of map fixed-pt. (which corresponds to our limit cycle) & if any $|\mu| > 1$ then instability. We didn't discuss case of $|\mu| = 1$ for some eigenvalue.

However, for this problem we are guaranteed to have one eigenvalue $\mu = 1$, i.e. $\exists v$ s.t. $Mv = v$. We can show this by finding v .



Let $v = \dot{x}(t) \Big|_{t=0}$ & show $Mv = v$

Proof: If $x(t)$ satisfies $\dot{x} = V(x)$, then so does $x(t+\tau)$ for any τ

$$\begin{aligned} \text{given } \dot{x}(t) &= V(x(t)) & x(t) &= x(t+\tau) \\ \Rightarrow \dot{x}(t+\tau) &= V(x(t+\tau)) \end{aligned}$$

$$\frac{d}{d\tau} [\dot{x}(t+\tau) = V(x(t+\tau))] \Big|_{\tau=0}$$

$$\frac{d}{dt} \dot{\gamma}(t) = \underbrace{D_x V(\gamma(t))}_A \dot{\gamma}(t)$$

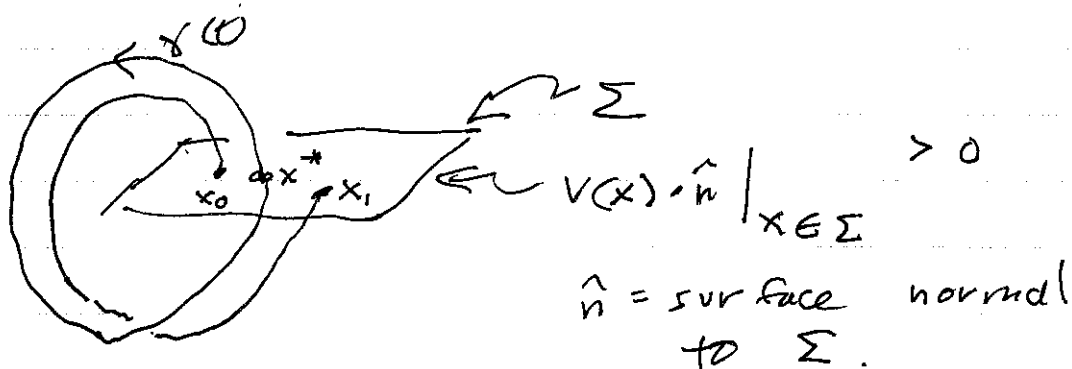
i.e. $\dot{\gamma}(t)$ satisfies $\dot{y} = A(t)y$ so

$$\dot{\gamma}(T) = M \dot{\gamma}(0)$$

But $\gamma(t)$ is T -periodic, so $\dot{\gamma}(t)$ is also T -periodic $\dot{\gamma}(T) = \dot{\gamma}(0) = M \dot{\gamma}(0) \Rightarrow \mu = 1$

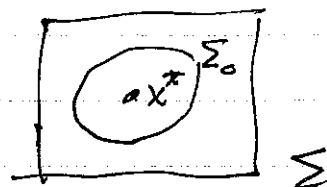
Approach 2 for computing linear stability of a limit cycle: Poincaré return map.

Introduce a "surface of section" Σ that's transverse to the flow near $\gamma(t)$ (perhaps only defined in some small region.)



Poincaré return map P

$$P: \Sigma_0 \rightarrow \Sigma, \quad \Sigma_0 \subset \Sigma$$



$P(x^*) = x^*$, where x^* is pt. on limit cycle $\gamma(t)$ that is in Σ_0

$$P(x_0) = t_1$$

$$P(x) = \bigcap_{I(x)} (x)$$

$T(x)$ = First return time to Σ
 $T(x) > 0$.

$P(x)$ is $(n-1)$ -dimensional map

$$x_{k+1} = P(x_k), \quad P(x^*) = x^*$$

linearize about $x = x^*$: $x_k = x^* + y_k$

$$\underline{x_{k+1}} = \underline{x^*} + y_{k+1} = P(x^* + y_k)$$

$$= \underline{P(x^*)} + D_x P(x^*) y_k + \dots$$

Linearized map:

$$y_{k+1} = D_x P(x^*) y_k$$

perturbations decay
 satisfy $|h| < 1$.

if eigenvaluesⁿ of $D_x P(x^*)$

Lecture 8 p.8

How are these $(n-1)$ eigenvalues related to Floquet multipliers of monodromy matrix M ?

Thm 4.55 in text (p.151)

$$\text{spec}(M) = \text{spec}(D\mathcal{P}(x^*)) \cup \{1\}$$

$\nearrow$
 n
 Floquet multipliers on $n \times n$ monodromy matrix

$\uparrow$
 $n-1$ eigenvalues of linearized $(n-1)$ -dim return map $\mathcal{P}$

$\nwarrow$
 associated with perturbations along δ

